# Handout 13: The concept check



Name: _____ Day 1 or Day 10 (circle one)

Ten questions about ideas, not about Python. Work alone, on paper, for 20 minutes. Nothing here is graded. On Day 1 nobody has seen Remus yet, so write what you think and leave a question blank if you have no idea; the same sheet comes back on Day 10, and the two together show how far you traveled.

**1.** A chess program has five parts: the board, the moves, the rules, the search, and the evaluation. Write one job for each part you can.

| Part | Its job |
|---|---|
| Board | |
| Moves | |
| Rules | |
| Search | |
| Evaluation | |

**2.** Remus stores the board as 8 rows of 8 squares, both counted from 0. Row 0 is rank 8 (the top of the board, where Black starts) and column 0 is file a. Which square name is row 6, column 4? Which (row, col) is h1?

**3.** A program first lists every square a piece could move to, then throws some of those moves away. Which moves does it throw away, and why?

**4.** It is White's turn. White can play move A or move B. After A, Black's two replies leave the score at +3 or +8 for White. After B, Black's two replies leave +5 or +1. Black always picks the reply that is worst for White. Which move should White choose, and what score should White expect?

```
          White to move
         /             \
        A               B
      /   \           /   \
    +3    +8        +5    +1
```

**5.** From the start position White has 20 moves, and after one move by each side there are 400 positions. Why would a chess program count positions like this? What would a count of 401 tell you?

**6.** In question 4, White has already looked at move A and knows it is worth +3. White now starts looking at move B and sees the reply that leaves +1. Does White need to look at B's other reply? Why or why not?

**7.** A queen counts 9, a rook 5, a knight or a bishop 3, and a pawn 1. White has a queen and two pawns. Black has two rooks and a knight. Who is ahead, and by how much?

**8.** A sample hides one hot spot. Three squares have been measured: the readings are below. Mark the square you would measure next, and say why in one sentence.

```
    a   b   c   d   e
  +---+---+---+---+---+
1 | 2 |   |   |   |   |
  +---+---+---+---+---+
2 |   |   |   |   |   |
  +---+---+---+---+---+
3 |   |   | 40|   |   |
  +---+---+---+---+---+
4 |   |   |   |   |   |
  +---+---+---+---+---+
5 |   |   |   |   | 0 |
  +---+---+---+---+---+
```

**9.** Every crystal has its own fingerprint of X-ray peaks, and a powder that mixes crystals shows all their peaks at once. Crystal A has peaks at 20, 30, and 45. Crystal B has peaks at 20 and 38. Crystal C has peaks at 30, 45, and 60.

- A powder shows peaks at 20, 30, 38, and 45. Which crystals are in it?
- Another powder shows peaks at 30 and 45 only. Which crystals are in it? If you cannot be sure, say what you would do next.

**10.** In one sentence each: what does a computer do the same way when it picks a chess move and when it picks the next measurement in a laboratory? What is different?